Surface Temperature Lapse Rates on the Juneau Icefield

I. Abstract

Surface temperature lapse rate (hereafter, lapse rate) is a key parameter in models of many surface processes, including in melt of snow and ice on glaciers. However, it is often assumed to be a fixed value, despite research demonstrating a large range of values in complex topography. This project leverages the ease of using small inexpensive temperature sensors to study lapse rates as part of the Juneau Icefield Research Program (JIRP), a glaciology field camp for undergraduates. Using an array of sensors, students design and conduct experiments to study the spatial variability of lapse rates across the icefield. Students use the data to understand the impact of lapse rates on mass balance, and thus glacier change, on the icefield. This proposal is to purchase additional temperature sensors to expand the research goals of the project. These goals fall into two initiatives: i) observation of the full melt season through year-round sensor deployments and ii) characterization of small-scale spatial variability with a dense grid of sensors. The proposed research will expand opportunities for undergraduate research and collaboration outside UW. It will also provide preliminary results for a NSF grant proposal and contribute to the next phase of my PhD research.

II. Background

On the Juneau Icefield Research Program (JIRP) students learn the fundamentals of glaciology, mountaineering skills, and gain research experience using field-based methods. Over the course of the summer, the program traverses the icefield between Juneau, AK and Atlin, BC, doing fieldwork from a series of bare-bones research stations interspersed across the icefield. In summer of 2022, Jess Badgeley (ESS alum) initiated this project as faculty on the program, studying the spatial variability of temperature lapse rates on the icefield. Last summer, Mira Berdahl (ESS post-doc) and I were both faculty for JIRP and worked with Jess to continue the project.

The project's charm comes from the choice of sensor: inexpensive coin-sized temperature sensors called iButtons. Relative to a weather station, iButtons are imprecise and more sensitive to environmental biases (e.g. solar forcing). However, they require no supporting infrastructure, and can be deployed at multiple field sites for a fraction of the price of a single station-grade sensor. The beauty of this approach is in measuring lapse rate rather than absolute temperature. For identical sensors exposed to the same local conditions, bias in absolute temperature is consistent, and so cancels out when calculating a lapse rate. Individually, each sensor has low resolution. But by combining observations over time, the relationship *between* sensors quickly converges to a precise value. With enough sensors, transects can be deployed over the same time period and the lapse rate at different locations can be compared.

For many students, this project is their first exposure to working with lapse rates. Naturally, they want to know the importance of measuring them. We have found the most intuitive way to make this connection is through the location of the freezing elevation or 0 °C isotherm. For a larger lapse rate (e.g. -10 °C km^{-1}), the elevation of the isotherm is low (i.e., the icefield stays cold even when the temperature at sea level is warm). But for smaller lapse rates (e.g. -4 °C km^{-1}), the isotherm rises, meaning that more glacier area is exposed to melt. Because the surface slope of the Juneau Icefield is very small (i.e., most of its area is at a similar elevation), a small increase in the isotherm elevation exposes a large area to melt. Currently, the isotherm elevation is below the majority of the icefield, and the melt area is small. But when the melt area increases, the icefield will lose volume much more quickly. Currently, even our best models fail to resolve weather at the glacier scale. In complex topography, the location of the isotherm depends as much on local climate as it does on the warming trend, meaning its elevation varies with local conditions. But by characterizing lapse rates at locations across the icefield, we can identify areas that are more susceptible to warming. This summer, part of student research will be modeling this relationship by mapping melt area on the icefield for different lapse rates, and comparing them to the “standard” lapse rate of -6.5 °C km^{-1}. For the different assumptions, students will then use a glacier model to simulate the impact of lapse rate on glacier change.

III. Proposed Work

This proposal is to fund the purchase of additional sensors and the associated materials for their deployment. These additional sensors will allow us to expand the scope of the project’s research, which will in turn increase opportunities for students. Broadly, the proposed work will help to better understand spatial variability in lapse rates and their effect on glacier change. This takes the form of two initiatives: i) observation of the full melt season through year-round sensor deployment, and ii) characterization of small-scale spatial variability with a dense grid of sensors.

Year-round deployments will allow us to collect data over the entire melt season. It will also allow deployment of sensors at more remote locations, towards the goal of instrumenting a transect across the maritime-continental transition. Southeast Alaska is perhaps unique in having maritime and continental glaciers that occur as part of the same continuous glacier complex. Due to the orographic effect, we expect glaciers to experience an increasingly continental climate towards the eastern (lee) side of the icefield. In the free atmosphere, the lapse rate of fully water-saturated air is -4 °C km^{-1}. In completely dry air, it is -10 °C km^{-1}. We hope to capture the change in lapse rate resulting from air drying as it passes over the icefield. By associating this data with melt rates, we can assess how maritime and continental glaciers will respond differently to changes in climate. Observing the lapse rate across this transition will give us insight into what effects dominate at the glacier surface.

While we expect such differences in lapse rate at the regional scale, we expect just as much variability in lapse rate at the local scale. Deploying a dense grid of sensors on a single glacier, we can study the effects of microclimate on melt rates, such as from topographic shading or katabatic winds. Instead of a lapse rate, with a grid of sensors we could calculate a lapse gradient, allowing for new analysis techniques. Aim to use the change in gradient over time to identify the effect of katabatic winds and solar shading on surface temperature. The time-averaged gradient could be used to characterize patterns of surface energy on the glacier. Analyzing these patterns for a single glacier would allow us to identify other glaciers where local processes may play a significant role.

Once collected, these datasets will expand our ability to use lapse rate as a teaching tool and open new possibilities for student research projects. However, collecting the data requires sensors to be deployed during the summer, which would leave none available for use on the student program. If funded, this proposal will provide the additional sensors required to collect these datasets concurrently while maintaining the existing project.

IV Future Work

A guiding principle for this project is to be a model of collaborative science. Because faculty join JIRP for different parts of the summer, collecting repeat data at some locations requires recruiting the help of another faculty. For faculty who specialize in modeling, JIRP's focus on field methods can make teaching difficult. This project allows faculty to collect data that can easily be tied to models of snow or glacier change. Similarly, use of these datasets is an accessible entry point for undergraduate research. The data from temperature sensors is easy to process and interpret, yet can be applied to a remarkable breadth of scientific questions. For the last two years, cohorts of students continued analysis following the summer and have presented work from this project at AGU. For both students and faculty, our goal is to create the infrastructure to encourage engagement in this project. Preparations for this summer will include instructional material on using the sensors, a procedure for their deployment, and premade analysis notebooks. By simplifying things like data management and processing, the focus can be shifted back to the science.

Both initiatives would make for extremely unique datasets and represent the direction I hope to go with my PhD research. As part of my own research, I plan to study the relationship between surface energy and lapse rates at the local scale, as well as the interaction between weather and glaciers at the regional scale. This project combines the regional and local scale to better understand the role glaciers play in their local climate and how that is likely to change over time. We plan to use the data from this summer as preliminary results for a NSF grant proposal to continue exploring lapse rates on glaciers.